

Preparing Medical Imaging Data for Machine Learning

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Artificial intelligence (AI) continues to garner substantial interest in medical imaging. The potential applications are vast and include the entirety of the medical imaging life cycle from image creation to diagnosis to outcome prediction. The chief obstacles to development and clinical implementation of AI algorithms include availability of sufficiently large, curated, and representative training data that includes expert labeling (eg, annotations). Current supervised AI methods require a curation process for data to optimally train, validate, and test algorithms. Currently, most research groups and industry have limited data access based on small sample sizes from small geographic areas. In addition, the preparation of data is a costly and time-intensive process, the results of which are algorithms with limited utility and poor generalization. In this article, the authors describe fundamental steps for preparing medical imaging data in AI algorithm development, explain current limitations to data curation, and explore new approaches to address the problem of data availability.

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Learning Objectives:

After reading the article and taking the test, the reader will be able to:

- List the different steps needed to prepare medical imaging data for development of machine learning models
- Discuss the new approaches that may help address data availability to machine learning research in the future
- Identify properly de-identified medical data according to the U.S. Health Insurance Portability and Accountability Act (HIPAA) and European General Data Protection Regulation (GDPR) standards

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Artificial intelligence (AI), as a field defined broadly by the engineering of computerized systems able to perform tasks that normally require human intelligence, has substantial potential in the medical imaging field (1). Machine learning and deep learning algorithms have been developed to improve workflows in radiology or to assist the radiologist by automating tasks such as lesion detection or medical imaging quantification. Workflow improvements include prioritizing worklists for radiologists (2,3), triaging screening mammograms (4), reducing or eliminating gadolinium-based contrast media for MRI (5,6), and reducing the radiation dose of CT imaging by advancing image noise reduction (7–9). Automatic lesion detection by using machine learning has been applied to many imaging modalities and includes detection of pneumothorax (10,11), intracranial hemorrhage (12), Alzheimer disease (13), and urinary stones (14). Automatic

quantification of medical images includes assessing skeletal maturity on pediatric hand radiographs (15), coronary calcium scoring on CT images (16), prostate classification at MRI (17), breast density at mammography (18), and ventricle segmentation at cardiac MRI (19,20). Yet substantial implementation and regulatory challenges have made application of AI models in clinical practice difficult and limited the potential of these advancements. Nearly all limitations can be attributed to one substantial problem: lack of available image data for training and testing of AI algorithms.

Currently, most research groups and companies have limited access to medical images, while the small sample sizes and lack of diverse geographic areas hinder the generalizability and accuracy of developed solutions (21). Although small data sets may be sufficient for training of AI algorithms in the research setting, large data sets

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Abbreviations

AI = artificial intelligence, DICOM = Digital Imaging and Communications in Medicine, PACS = picture archiving and communication system

Summary

Supervised artificial intelligence (AI) methods for evaluation of medical images require a curation process for data to optimally train, validate, and test algorithms. The chief obstacles to development and clinical implementation of AI algorithms include availability of sufficiently large, curated, and representative training data that includes expert labeling (eg, annotations).

Essentials

- Image data availability is an important hurdle for implementation of artificial intelligence (AI) in the clinical setting.
- AI researchers need to be aware of the data source and potential biases, which may affect generalizability of AI algorithms.
- New approaches such as federated learning, interactive reporting, and synoptic reporting may help to address data availability in the future; however, curating and annotating data, as well as computational requirements, are substantial barriers.

with high-quality images and annotations are still essential for supervised training, validation, and testing of commercial AI algorithms. This is especially true in the clinical setting and is well outlined by Park and Han (22).

Most health care systems are not adequately equipped to share large amounts of medical images. Even when development is possible, medical data are often stored in disparate silos, which is not optimal for medical AI development that can be broadly used in clinical practice(s). Furthermore, simply achieving access to large quantities of image data is insufficient to allay these shortcomings. Adequate curation, analysis, labeling, and clinical application are critical to achieving high-impact clinically meaningful AI algorithms. We describe a process of labeling, curating, and sharing medical image data for AI algorithm development, followed by an in-depth discussion of alternative strategies to achieve responsible data sharing and applications in AI algorithm development for optimal clinical impact. To date, to our knowledge, this is the first work that gives an overview of the process of medical imaging data preparation for machine learning.

Conflicts of Interest

Data and information were controlled by authors who are not industry employees. Two authors (W.A.K. and C.H.) are industry employees of Segmed (Palo Alto, Calif), a company that delivers machine learning training data for medical imaging. Four authors (M.J.W., J.W., H.H., and M.P.L.) are advisors and stockholders of the same company. However, for this project none of the authors received financial or research support from the industry and the current project itself also was not funded.

Data Preparation Overview

Before medical images can be used for the development of an AI algorithm, certain steps need to be taken. Typically, approval from the local ethical committee is required before medical data may be used for development of a research or a commercial AI algorithm. An institutional review board needs to evaluate the risks and benefits of the study to the patients. In many cases existing data are used, which requires a retrospective study. Because the patients in this type of study do not need to undergo any additional procedures, explicit informed consent is generally waived. With clinical trials, each primary investigator may need to provide approval to share data on their participants. In case of a prospective study, where study data are gathered prospectively, informed consent is necessary. After ethical approval, relevant data needs to be accessed, queried, properly de-identified, and securely stored. Any protected health information needs to be removed both from the Digital Imaging and Communications in Medicine (DICOM) metadata, as well as from the images (23). If the data are intended for open-source research efforts, then additional human inspection of each image is standard because some images contain free-form annotations that have been scanned and cannot be removed reliably with automated methods. The quality and amount of the images vary with the target task and domain. The next step is to structure the data in homogenized and machine-readable formats (24). The last step is to link the images to ground-truth information, which can be one or more labels, segmentations, or electronic phenotype (eg, biopsy or laboratory results). The entire process to prepare medical images for AI development is summarized in Figure 1.

Accessing and Querying Data

Developers of AI algorithms are typically not located within a hospital and therefore often do not have direct access to medical imaging data through the picture archiving and communication system (PACS), especially when AI researchers are developing

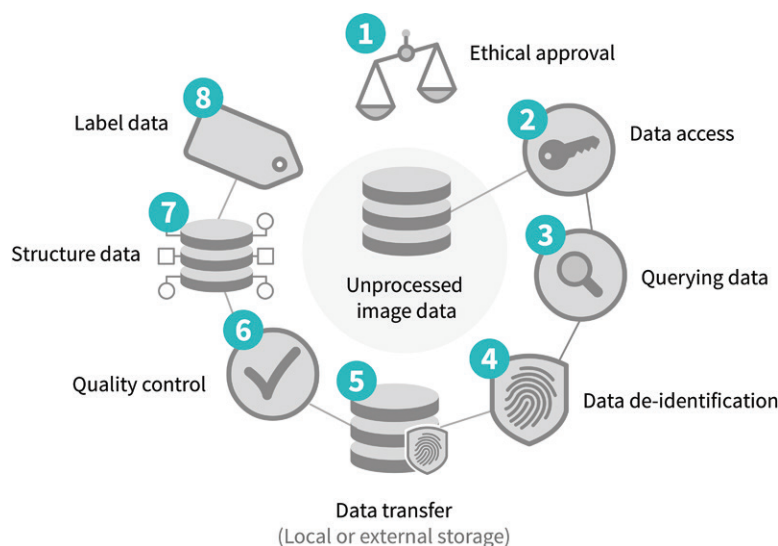


Figure 1: Diagram shows process of medical image data handling.

Table 1: Protected Health Information Identifiers according to the Health Insurance Portability and Accountability Act

Identifier
Name
Address*
All elements (except years) of dates related to an individual†
Telephone numbers
Fax number
E-mail address
Social Security number
Medical record number
Health plan beneficiary number
Account number
Certificate or license number
Any vehicle or other device serial number
Device identifiers and serial numbers
Web URL
Internet Protocol (IP) address
Finger or voice print
Photographic image‡
Any other characteristic that could uniquely identify the individual

Source.—Reference 101.

* All geographic subdivisions smaller than state, including street address, city, county, and zip code.

† Including birth date, admission date, discharge date, date of death, and exact age if older than 89 years.

‡ Photographic images are not limited to images of the face.

commercial algorithms. Access to PACS environments is limited to accredited professionals such as physicians, technologists, PACS managers, and clinical scientists. Making data accessible to AI developers is challenging and requires multiple steps, including de-identification of data (described later). The ideal approach is collaboration between clinicians and AI developers, either in-house or through collaborative research agreements.

Once data are accessible to AI developers, different strategies are available to search for medical images and clinical data. Custom search queries may, for example, consist of strings, international classification of disease codes, and current procedural terminology codes. Data can be systematically searched and extracted from hospital PACS and electronic medical records by using PACS or radiology information system search engines. For example, many PACS vendors allow user access to metadata such as annotations, creator, series and image number, and unique target lesion names and relations. These data can be exported in some PACS and further managed by other systems such as electronic medical records, cancer databases, and oncologist or other provider databases (25). Alternatively, software packages are available to simplify the process of data querying (26–28).

De-Identification

Although written informed consent from patients is not always necessary, according to the U.S. Health Insurance Portability

and Accountability Act, or HIPAA, and the European General Data Protection Regulation, both retrospectively and prospectively gathered data require proper de-identification. Sensitive information includes but is not limited to name, medical record number, and date of birth. A complete list of the 18 HIPAA identifiers is shown in Table 1. Identifiable information is commonly present in the DICOM metadata (header) and multiple tools are available to automatically remove this information (29). DICOM de-identification profiles are defined for a range of applications and used as the basis for de-identification workflows implemented in the Radiological Society of North America Clinical Trial Processor and the Cancer Imaging Archive (30). Besides the DICOM metadata, protected health information may also be embedded in images, which is often the case with US examinations or radiographs that are scanned into a health care system. Removal of embedded information requires more advanced de-identification methods such as optical character recognition (31) and human review for handwriting on scanned images not always recognized by automated methods. Care must also be taken not to inadvertently mix data sets, because doing so increases the individual risk of reidentification through cross-linking of nonrelated data points (32). Finally, medical data can be anonymized with *k*-anonymity, which transforms an original data set containing protected health information to prevent potential intruders from determining the patient's identity (33). For posting radiology data in open-source research efforts, the DICOM metadata is often removed completely or converted to another format such as Neuroimaging Informatics Technology Initiative, or NIFTI, which retains only voxel size and patient position. Totally removing the DICOM metadata for open-source research efforts prevents privacy issues but reduces the value of data, because metadata is important for AI algorithm development.

Important protected health information that can be potentially overlooked, yet can act as “identity signatures,” include the HIPAA items full-face photos and comparable images, as well as biometric identifiers (ie, retinal scan and fingerprints). For example, head and neck CT data can qualify as comparable images. With widespread volumetric acquisition and ease of three-dimensional reformatting, the soft-tissue kernels or filters allow facial reconstruction that can identify the patient. Until there is a secure digital encryption method to alter identification without compromising clinical information, those making data publicly available need to take potential biometric signatures into consideration.

Data Storage

Data are commonly transferred to either a local data storage (single-center study) or an external data storage (multicenter study or commercial AI development). Data are usually stored at an on-premise server; however, with current cloud-based developments, data are increasingly stored in the cloud. Advantages of on-premise data storage include data safety and availability, but the potential of sharing data with other institutions is limited. Cloud-based data storage, on the other hand, is be-

coming more secure, improves the possibilities of sharing data, and provides data backup. Disadvantages of cloud-based storage include costs and the need for a fast internet connection.

Resampling Medical Images

Image perception of medical image data are relatively complex compared with nonmedical image perception tasks. Most convolutional neural networks for classification of images are trained and tested on two-dimensional images with fewer than 300×300 pixels (34). Medical images, however, exceed these dimensions; the in-plane spatial resolution is generally higher than 300×300 pixels, and many medical image studies are three-dimensional instead of two-dimensional. Training convolutional neural networks with images larger than 300×300 pixels is possible; however, computers with strong computational power are necessary. This problem is most relevant in high-resolution applications; examples include CT of the inner ear or full-field digital mammography. Solutions include downsampling of the image resolution or patch-based evaluation of only image parts with relevant information (eg, focus on the aortic region in an algorithm developed for aortic dissection segmentation). However, patch-based methods frequently have high computational demands and are time consuming to train. Model training can also be simplified by classifying labels to healthy (scale 0) and diseased at different levels; for example, from less severely diseased (scale 2) to more severely diseased (scale 4) (35).

Besides DICOM files, which contain metadata and image slices, other file types are also available. AI development with raw MRI or CT data (before images are reconstructed) is gaining interest and has a potentially valuable role. Advantages include an increased amount of information captured in raw data, and disadvantages include the large storage space needed and difficult interpretation of raw data without reconstructed images.

Choosing Appropriate Label and Ground Truth Definition

Current AI algorithms for medical image classification tasks are generally based on a supervised learning approach. This means that before an AI algorithm can be trained and tested, the ground truth needs to be defined and linked to the image. The term *ground truth* typically refers to information acquired from direct observation (such as biopsy or laboratory results). Image labels are annotations performed by medical experts such as radiologists. These annotations can be considered ground truth if imaging is the reference standard (eg, pneumothorax). Choosing the appropriate label for a given imaging AI application requires a balance between finding the best discriminating categories (ie, normal vs emergent) and clinically relevant granularity (ie, subtype of liver lesion) depending on the desired task. With the exception of AI methods that enhance image quality, medical images in isolation are generally not suitable for developing diagnostic AI models unless associated with a diagnosis through the free-text radiology report (which require additional labeling strategies discussed below), expert consensus, segmentation, or an applied ground truth label such as electronic phenotyping (1).

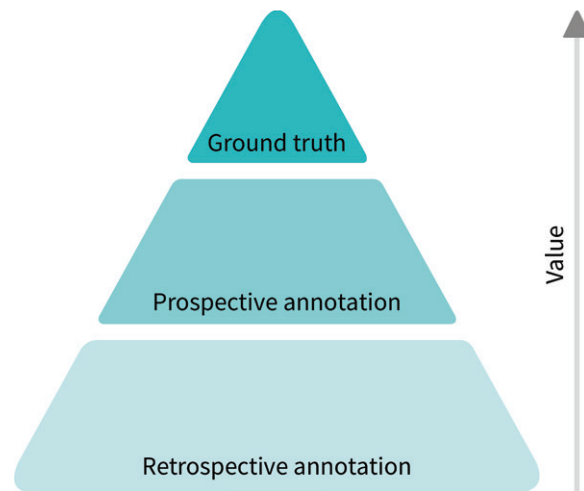


Figure 2: Diagram shows value hierarchy of imaging annotation. Most useful but least abundant is ground truth (pathologic, genomic, or clinical outcome data). Prospective annotation is incredibly valuable due to availability of contemporaneous information (clinical and/or laboratory data). By comparison, retrospective annotations are least valuable.

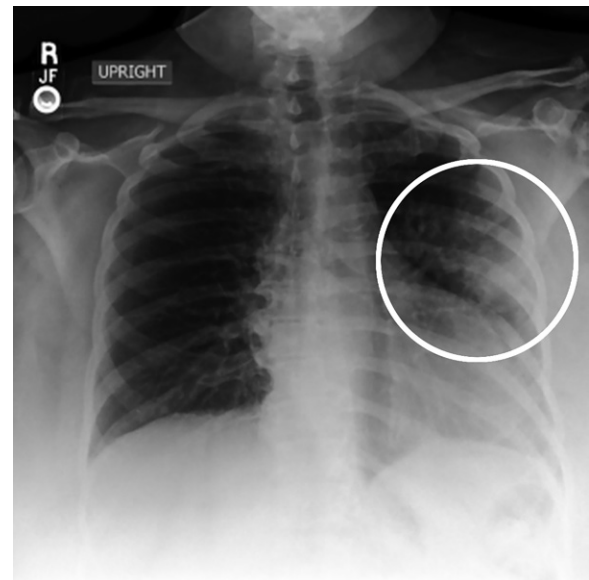


Figure 3: Image in posterior-anterior direction shows nonspecific abnormality on chest radiograph. Application of most accurate label for nonspecific finding such as opacity in left lung (circle) is challenging in absence of other clinical and laboratory data.

Although extracting structured labels from the radiology report text by using natural language processing may be ultimately the most scalable approach, researchers need to be cautious of the error rates both in the natural language processing techniques and the original text reports. In large quantities, it is known that AI algorithms can be trained on relatively low-quality data, but knowing the true ground truth for a given task to correlate with the imaging findings is the ideal (Fig 2). Medical imaging alone is considered ground truth for certain diagnoses, including intracranial hemorrhage, fractures, renal stone, and aortic dissection. However, the majority of findings is not definitive on the basis of imaging examinations alone.

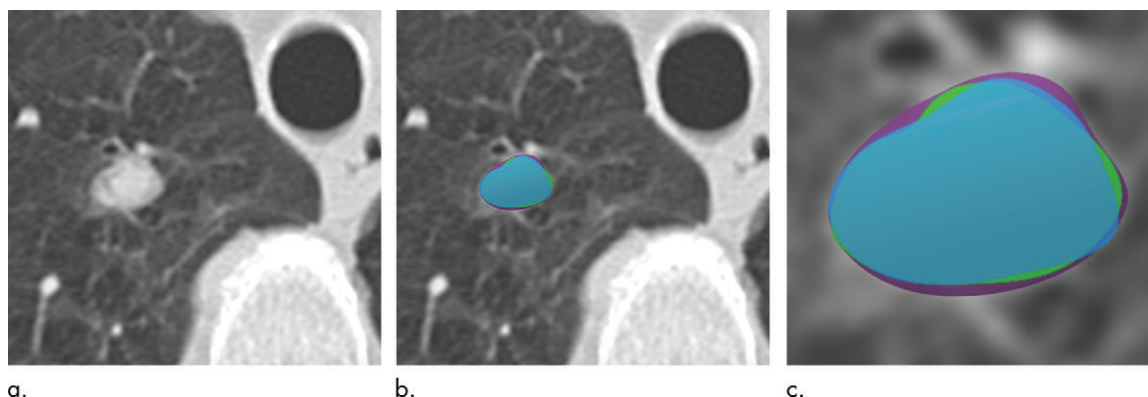


Figure 4: Axial images show medical image segmentations performed by experts. **(a)** CT examination of patient with lung nodule. **(b)** Nodule is independently and blindly segmented by three medical experts with free open-source software package (Horos, version 3.3.5; Nimble d/b/a Purview, Annapolis, Md). **(c)** Magnified image of segmentations. There are differences between segmentations; however, these differences are small and not clinically relevant.

and requires further follow-up, pathologic diagnosis, or clinical outcomes to achieve ground truth (ie, lung cancer, liver mass, pneumonia, etc). For example, an opacity on a chest radiograph has an extensive differential. It is difficult to know the ground truth without obtaining surgical, pathologic, genomic, or clinical outcome data (Fig 3). There are also situations in which one modality may support a diagnosis but require definitive confirmation by using another modality. For example, a head CT might have findings supporting a diagnosis of stroke, but an MRI could definitively confirm. Depending on the ultimate task or purpose of the AI algorithm, ground truth definition may require confirmatory clinical labeling beyond the radiology opinion or report such as a pathologic or surgical report, clinical outcome, or both. Accumulation of this clinical information for a large number of patients can be resource intensive and is often referred to as electronic phenotyping (36). Querying of nonimaging data such as clinical outcomes and patient demographics can often not be performed through PACS, but requires extraction of information from electronic medical records.

In general, imaging data can be labeled in a variety of ways including structured label(s), image annotations, image segmentations, and/or electronic phenotypes (1,37). More often, application of the imaging diagnosis based on expert interpretation or a consensus of experts based on a reinterpretation of the images, or free-text report is used (10). Another approach to labeling is through the use of segmentations including, for example, outlining lung nodules at CT of the chest (Fig 4).

Ground Truth or Label Quality

Accurate ground truth definition or image labels for a large number of radiology examinations are required to build accurate medical imaging AI models (38,39). There are guidelines for reporting diagnostic imaging aiming toward structured reporting, which would immensely reduce the effort needed to extract useful imaging labels. At present, however, the overwhelming majority of reports remain composed of free text (40). Novel semantic reporting systems that aim to index and codify free-text reports in real time are being developed, but are currently not widely available for large populations. As a

result, most centers attempting to use retrospective data are faced with large volumes of imaging studies and narrative reports that require substantial effort to label. Currently, there are many approaches to perform retrospective labeling, ranging from simple manual labeling by radiologists to automated approaches that can extract structured information from the radiology report and/or electronic medical record (41).

Outside of medical applications, manual labeling is a commonly used approach in acquiring labeled imaging data for AI applications. Large corporations often hire nonexperts to hand review and label large amounts of data needed to support automated services such as ranking web search results, providing recommendations, or displaying relevant ads (42,43). This approach can be effective in medical imaging as well, but is impractical in most cases when used on large populations because it is extremely time consuming (and costly) to use medical experts, particularly for advanced modalities such as CT, PET, or MRI. When a relatively small number of images are needed for AI development, medical expert labeling and segmentation may be feasible. Segmentation performance can be evaluated by using either the Dice coefficient or the more advanced simultaneous truth and performance level estimation, or STAPLE, algorithm (44). The STAPLE algorithm compares segmentations and computes a probabilistic estimate of the true segmentation. For narrowly focused applications such as colonic polyp classification and kidney segmentation, crowdsourcing of labels by nonexperts may be feasible (45,46). Heim et al (47) compared segmentations of the liver performed by nonexperts, engineers with domain knowledge, medical students, and radiologists. Despite the finding that the crowd needed more time, accuracy was similar between these groups. Crowdsourcing challenges include inaccuracy with anatomic variations and pathologies, quality control, and ethical issues such as sharing medical images with the crowd. Crowd-sourced labeling is mostly performed with web-based tools, which are freely available (48,49).

One solution is to extract information from the report of imaging findings through rule-based natural language processing (50,51) or recurrent neural networks (52,53). One of the most useful natural language processing methods is called topic modeling, which summarizes a data set with a large amount

of text to obtain gross insight over the data set. This approach characterizes document content based on key terms and estimates topics contained within documents. For example, documents associated with “brain MRI” would comprise key terms such as *axial*, *contrast*, *MRI*, *sagittal*, *brain*, *enhancement*, et cetera. Another class of architectures, recurrent neural networks, are neural network–based models that can be trained on a small sample of reports and rapidly achieve performance levels of the state-of-the-art more traditional natural language processing tools (54,55). Recurrent neural networks represent an important improvement to language modeling because a dependency of a word in narrative language can occur long distances apart, such as “No evidence for acute or subacute infarction.” In this example, the “No” is far from the target “infarction.” It can be confusing for traditional natural language processing tools but picked up with recurrent neural networks (54). As a result, strategies for extracting structured labels from unstructured text have emerged that have shown a great deal of promise for limited applications to apply structured labels in large populations and generate large labeled data sets of imaging studies (2,56).

Because radiology reports are most often unstructured and not created specifically for the development of AI algorithms, the extracted information contains noise (ie, has a relatively low quality). Neural networks can still be relatively robust when trained with noisy labels (57). However, one should be careful when using noisy labels for the development of clinically applicable algorithms because every labeling error could be translated to a decrease in algorithm accuracy. It is estimated that 2%–20% of radiology reports contain demonstrable errors (58).

Lastly, there is a trend toward interactive reporting where the radiologist report contains hypertext directly connected to image annotations (59). Such annotations have been used effectively for labeling of open-source data sets (60). Measurements can be performed in advance of radiologists by radiology preprocessors that improve annotation quality while saving radiologists time (61). Preliminary work on prospective labeling is showing that two-diameter measurements and ovals are better than one-diameter measurements and much better than arrows (62). In addition to structured reporting, to the level of synoptic reporting, this should contribute significantly to increased prospective expert-labeled data. Interactive reporting is becoming more common where radiologists routinely label images in three dimensions and connect directly to hypertext descriptions in their report. This may be a potential solution to the local labeling issue with research just beginning. Nevertheless, substantial collaborative efforts may ultimately be needed to arrive at widely adopted reporting and standardization of labeling of imaging studies such that interoperability of data sets and subsequent models is possible.

Data Sets

Development of AI algorithms by using supervised learning requires large and heterogeneous training, validation, and testing data sets.

Data Set Types

Similar to conventional regression modeling, AI models are trained by inputting medical images linked to ground truth outcome variables (eg, pneumothorax). Generally, the training imaging data set is larger than the validation and testing data sets in ratios of 80:10:10 or 70:15:15. To ensure generalizability of the AI algorithm, bias of the training data set should be limited. If an AI algorithm is trained with images from a European institution and the algorithm is used in an Asian population, then performance may be affected by population or disease prevalence bias. Similarly, if all the imaging training data were acquired by using one kind of imaging machine, it may not work as well on machines from other manufacturers, known as vendor or single-source bias. It is thus advised to use images from multiple diverse sources, or at least images representing the target population or health system in which the algorithm is to be deployed. After the algorithm is trained, a validation data set is needed to fine-tune the algorithm hyperparameters and to check for overfitting. Note that validation in AI algorithm development has a different meaning than in conventional statistical modeling. Here, validation means tuning of the algorithm until the final performance of the model is evaluated with a testing data set. Multiple internal validation methods are available; however, independent validation in an external data set is preferred over internal validation to properly evaluate generalizability (63). Even if an electronic phenotype is available (eg, biopsy results of a lung nodule), annotations are needed for training and validation data sets to inform the algorithm of the location of the specific lung nodule to allow the algorithm to better understand the images. The testing data set functions as the reference standard and is used to evaluate the performance of the algorithm. In multiple conditions, imaging is the reference standard (eg, pneumothorax), where high-quality annotations are needed for the testing imaging data set because this data set functions as the reference standard. The quality and veracity of the testing data set is arguably more important than that of the training set because this data set is used for performance testing and regulatory approval.

Data Set Size

To ensure generalizability, large training data sets are often essential. For specific targeted applications or populations, relatively small data sets (hundreds of cases) may be sufficient. Large sample sizes are especially required in populations with substantial heterogeneity or when differences between imaging phenotypes are subtle (35). The algorithm performance for computer vision tasks increases logarithmically with increased training data volume (64,65). Therefore, a proper sample size is needed. The main questions for the power calculation include the following: (a) which cases need to be included in the sample to allow for generalizability in a larger population, and (b) how many cases are needed to show an effect (66). The sample size calculation for test data sets should use traditional power calculation methods to estimate the sample size. In general, the development of generalizable AI algorithms in medical imaging requires statistically powered data sets in the order of

Table 2: Large Open-Source Medical Imaging Data Sets

Data Set Description	Image Types	No. of Patients	Ground Truth	Single or Multiple Institutions
American College of Radiology Imaging Network National CT Colonography Trial (ACRIN 6664) (102)	CT	825	Pathology (biopsies)	Multiple
Alzheimer's Disease Neuroimaging Initiative (103)	MRI, PET	>1700	Clinical (follow-up)	Multiple
Curated Breast Imaging Subset of the Digital Database for Screening Mammography (36)	Mammography	6671	Pathology (biopsies)	Multiple
ChestX-ray8, National Institutes of Health chest x-ray database (41)	Radiography	30 805	Imaging reports	Single
CheXpert, chest radiographs (79)	Radiography	65 240	Imaging reports	Single
Collaborative Informatics and Neuroimaging Suite (104)	MRI		Clinical (follow-up)	Multiple
DeepLesion, body CT (60)	CT	4427	Imaging	Single
Head and neck PET/CT (105)	PET/CT, CT	298	Pathology (biopsies), clinical (follow-up)	Multiple
Lung Image Database Consortium image collection (106)	CT, radiography	1010	Imaging, clinical for a subset	Multiple
MRNet, knee MRI (80)	MRI	1370	Imaging reports	Single
Musculoskeletal bone radiographs, or MURA (107)	Radiography	14 863	Imaging reports	Single
National Lung Screening Trial (108)	CT, pathology	26 254	Clinical (follow-up)	Multiple
PROSTATEx Challenge, SPIE-AAPM-NCI Prostate MR Classification Challenge (109)	MRI	346	Pathology (biopsies), imaging	Multiple
Radiological Society of North America Intracranial Hemorrhage Detection (110)	CT	25 000	Imaging	Multiple
Cancer Genome Atlas Kidney Renal Clear Cell Carcinoma data collection (111)	CT, MRI	267	Pathology (biopsies), clinical (follow-up)	Multiple
Virtual Imaging Clinical Trial for Regulatory Evaluation (112)	Mammography, digital breast tomosynthesis	2994	Imaging	Multiple

Note.—AAPM = American Association of Physicists in Medicine, NCI = National Cancer Institute, SPIE = Society of Photo-Optical Instrumentation Engineers.

hundreds of thousands or millions, which is problematic for many researchers and developers.

One partial solution for this problem may be semisupervised learning. Fully annotated data sets are needed for supervised learning, whereas semisupervised learning uses a combination of annotated and unannotated images to train an algorithm (67,68). Semisupervised learning may allow for a limited number of annotated cases; however, large data sets of unannotated images are still needed.

Another potential future solution to increase data sample size may be the generation of synthetic data through generative adversarial networks (69). Generative adversarial networks have the potential to synthesize unlimited numbers of high-quality realistic images that can be added to training data sets for development of detection and classification algorithms. First results in synthesized radiographs and mammograms are promising. However, limited evidence is available, especially when abnormalities are present on images (69,70).

Data Sources

Most academically developed AI algorithms in medical imaging have been trained, validated, and tested with local data from a single institution (1). Whereas multi-institutional data

from different geographic areas would include a wide variety of imaging machines, ethnicities, and pathologies, single-institutional data are commonly used due to lack of access to multi-institutional data. Many medical centers lack motivation and resources to share data with other institutions or companies that develop AI algorithms due to regulatory and privacy issues, although medical image data can be shared without violating General Data Protection Regulation or HIPAA regulations with proper de-identification methods and secure data handling. Currently, medical image data are stored in isolated decentralized silos, limiting the development of generalizable unbiased AI algorithms, which could theoretically be solved by having centralized data storage systems. When data are being made available to AI developers, appropriate data management is essential. Wilkinson et al (71) describe the FAIR (findability, accessibility, interoperability, and reusability) principle for good data management.

Open-Source Data Sets

An increasing number of data sets has been open sourced to address the problem of data access in medical research. Data sets are available in a wide range of domains from neuroimaging (72–77), breast imaging (36,78), chest radiographs (41,79),

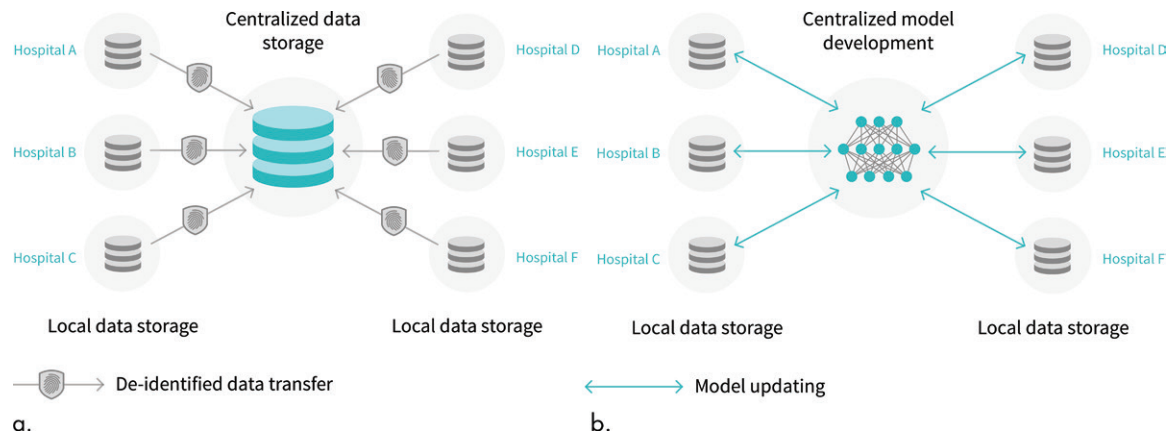


Figure 5: Diagram shows centralized versus federated learning. **(a)** Current artificial intelligence (AI) model development is through centralized model, in which de-identified data are transferred to centralized data storage system where AI algorithm can be developed. **(b)** In the future, federated learning may be used, in which data stays in each hospital. With federated learning, instead of transferring data outside each hospital, data stays in hospitals and AI model is sent to and trained in hospitals.

knee MRI (80), body CT (60), and others; a list of well-known open-source data sets is given in Table 2. Whereas open-source data sets stimulate the development of novel AI algorithms in the medical imaging field, there are important limitations. First, there is a wide variety of number and quality of images and availability of metadata and clinical information. Second, some open-source data sets are (partly) acquired by using outdated machines, contain low-quality images, lack expert labeling or data curation, or have a sample size that is too small to reach high-quality algorithms that can be used clinically. Moreover, many open-source data sets are restricted to noncommercial (research only) use (79). This is a major limitation for researchers wishing to develop marketable algorithms, as commercial adoption is a common avenue for clinical deployment.

Bias

One of the most important limitations of training AI algorithms based on data from a single institution or from multiple institutions in a small geographic area is sampling bias. If an AI algorithm trained this way is applied to a different geographic area, then results of the algorithm may be unreliable due to differences between the sample population and target population (81). Other sources of bias include differences in age, proportions of race and sex, use of imaging machines (vendors, types, acquisition protocols), and prevalence of diseases. There may even be biases that researchers are unaware of, such as variations in local practice. For many medical applications there is a substantial variability between experts who evaluate images, which is true in clinical practice because it is inherent when labels and segmentations are manually created. This variability may result in biased labels and segmentations that may be mitigated by having multiple experts evaluating the same case (82,83). However, substantial costs and time delays often limit image assessment by multiple experts. Another important reason for using training data from a widespread geographic region is the availability of AI algorithms to patients in developing countries. Most AI algorithms are developed by research groups and companies

who have access to training data sets with images from patients in developed countries (84).

Data Format

The two key types of formats relevant to AI application development are image data formats and image annotation formats. Nearly all PACS store medical images in DICOM format, which is the international standard for image objects. However, groups who collect images may convert them from DICOM to other formats such as portable networks graphics, or PNG, tagged image file format, or TIFF, or NIFTI for ease of distribution. However, one should keep in mind that important DICOM metadata are removed with these conversions. Image converting programs are sufficiently prevalent and accessible that there is usually no problem accessing and using image data acquired from multiple institutions in AI application development.

Unlike with image data, image annotations are not stored in a single common format. A major limitation of current commercial imaging systems that acquire image annotations (eg, for tracking cancer lesions [85–91]) is that they generally do not store annotations in a format that permits reuse for AI development. Image annotations are commonly stored in PACS and other systems as DICOM presentation state objects (92), which often vary among vendors and from which it is difficult to extract regions of interest, and usually these objects do not contain image labels. Even if they use DICOM structured reporting, or DICOM-SR (93), which provides different use case–specific templates for storing explicit details of image annotations, similar kinds of annotation data across systems may be stored by using different types of DICOM SR templates that thwart interoperability and reuse of annotations for AI development when acquiring them from different sites or even from different commercial systems within a single site. An important image annotation format for saving regions of interest is the DICOM segmentation map format (92,94), which is part of the DICOM standard. For nongraphic annotations, namely image labels such as radiologic findings or diagnoses, the annotation and image

markup, or AIM, format (95–97) was developed and recently incorporated into the DICOM-SR standard (92). A few vendors of AI products and PACS have begun supporting this standard, as well as the open-source ePAD web-based image viewing and annotation tool (48). Adopting these standards for storing these image annotation data will enable multicenter sharing, aggregating, and repurposing of image data for studying new quantitative imaging biomarkers.

Federated Learning

In 2017, Google (Mountain View, Calif) introduced federated learning, a potential solution for the availability of data for AI algorithm development (98). With current practice, de-identified data are transferred from the hospital (or silo) to a central storage system, whereas with federated learning the data stay in the hospital while the algorithm can be trained locally at multiple locations (Fig 5). Moreover, the algorithm itself takes up substantially less storage compared with image data. Therefore, distribution of algorithm training across institutions may be a viable solution.

Different approaches have been proposed, including parallel and nonparallel training methods. Parallel training methods were developed to speed up algorithm training by splitting the data set in separate samples. Different models are trained on each split of the data and finally the gradients are transferred to a central model (99,100). With nonparallel training, a sequential or cyclical method is used where the model is updated with data from each institution. Federated learning has not yet been evaluated extensively in the medical imaging field. Chang et al (35) simulated image classification with nonparallel federated learning across four institutions and found better performance than with models based on single institutions and similar performance as centrally hosted patient data.

Despite the potential benefits of this technique, there are important problems that need to be solved before federated learning can be applied. First, scalability is limited because image annotation and labeling needs to be performed to an agreed standard per site, because data cannot leave an institution. Second, substantial computation resources may need to be replicated and placed within each facility for federated learning. Third, preprocessing and organizing the data for ingestion by the algorithm (an estimated 80% or more of the effort) is challenging in the federated approach, because the visibility of data to the algorithm developers is impeded. Fourth, variations in terminology across sites requires mapping to a common controlled terminology. Fifth, only gradient information is shared with algorithm developers, which is a step toward protecting protected health information data leaving the hospital because raw image data does not have to be shared. However, sensitive information may still be present in the gradient information. Lastly, there will be heterogeneity across different institutions in terms of patient populations, data volume, data format, et cetera. However, although federated learning is an attractive model, it is not yet applicable in the clinical setting due to its very early phase. Federated solutions to data structuring, labeling, and computing—as well as agreed cross-site standardization of all data formats—will need to be developed for this approach to achieve large-scale adoption.

Data Label Relationship to Future Implementation

It may never be possible to constrain medical imaging to a finite number of labels deterministically. While most of the AI research and solutions in medical imaging today are still carried out solving specific isolated tasks and based on curated data labeling, this is an approach at odds with the desired future state of a continuous learning environment enabling the autonomous incremental adaption to an ever more complex medical system. Practically, this will require the infrastructure to update the prediction model to take into account different data distributions or new information. Data curation and labeling strategies will therefore adapt to new AI techniques continuously learning from streaming (even multimodal) data, which will challenge any static approach to data labeling and training.

Conclusion

Image data availability is an important hurdle for implementation of artificial intelligence (AI) in the clinical setting. AI researchers need to be aware of the data source and potential biases, which may affect generalizability of AI algorithms. New approaches such as federated learning, interactive reporting, and synoptic reporting may help to address data availability in the future. However, curating and annotating data, as well as computational requirements, are substantial barriers.

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